Deep Learning-based Pneumonia Detection Using Chest X-ray Images

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Abstract

This study evaluates the ResNet-18 architecture for automated pneumonia detection in pediatric chest X-rays, comparing random initialization against transfer learning. To address significant class imbalances and an insufficient original validation set, the data was re-split and trained using a Weighted Random Sampler, along with robust data augmentation to prevent overfitting to non-clinical image artifacts. Results indicate that although the scratch-initialized model achieved high test accuracy, it showed a strong bias toward the pneumonia class, correctly identifying only 70.37% of normal cases.In contrast, the fine-tuned transfer learning model demonstrated superior clinical utility, achieving a more balanced performance with 88.03% accuracy for Normal cases and 92.16% for Pneumonia. Grad-CAM [9] visualizations revealed that the scratch model's failures often stemmed from misinterpreting peripheral rib structures and diaphragmatic edges as pathological opacities. These findings underscore that transfer learning, combined with targeted data preprocessing and interpretability tools, remains the most effective strategy for pediatric diagnostics in data-constrained medical environments.

The source code for this project is available at: `https://github.com/olabamipetaiwo/Pneumonia-Clssification`

1. Introduction

Pneumonia remains a significant clinical challenge and is consistently identified as the leading cause of childhood mortality, resulting in approximately 2 million deaths among children under five each year, according to the World Health Organization. The classification of medical images is challenging due to high inter-observer variability and the subtle visual cues required to distinguish among pathologies. Deep learning methods are particularly well-suited to this task, as they automatically extract complex, hierarchical features from raw pixel data, thereby eliminating reliance on manually defined markers that may not capture the nuanced textures and spatial relationships essential for clinical diagnosis. The Chest X-Ray Images (Pneumonia) dataset, obtained from Kaggle, comprises 5,863 anterior-posterior chest X-ray images of pediatric patients aged one to five years from the Guangzhou Women and Children's Medical Center. The dataset is organized into three directories: train, test, and validation, each with subfolders for the two target categories, Pneumonia and Normal. The dataset demonstrates significant class imbalance, with the Pneumonia category containing substantially more images than the Normal category, reflecting the clinical distribution observed in the retrospective cohort. Addressing this class imbalance is a critical step in data preprocessing. Image resizing standardizes input dimensions for the neural network, and normalization stabilizes the learning process. Furthermore, data augmentation techniques, including random rotation and cropping, are employed to reduce overfitting and improve the model's generalizability, despite the disproportionate representation of pneumonia cases. The primary objective of this study is to compare binary classification using the ResNet-18 architecture. The model is first trained from scratch with random parameter initialization, getting training tricks from [2], and subsequently fine-tuned on the target X-ray training set using a ResNet-18 model pre-trained on ImageNet. This methodology aims to maximize testing accuracy and to evaluate performance differences between these two training paradigms, thereby determining the most effective approach for pediatric pneumonia detection.

2. Methodology

2.1. Network Architecture

The ResNet-18 architecture is organized into four sequential stages, each comprising two residual blocks, preceded by an initial convolutional layer with a stride and a max-pooling operation. Within each residual block, two convolutional layers are utilized, followed by batch normal-

ization and Rectified Linear Unit (ReLU) activation functions. The defining characteristic of this architecture is the shortcut or identity connection, which adds the block's input element-wise to its output. This design enables the layers to learn an additive residual function rather than a direct underlying mapping, which significantly simplifies network optimization by providing a direct path for gradient backpropagation during training.

For this medical task, the standard architecture was modified for binary classification between Normal and Pneumonia. After the final convolutional stage, global average pooling reduces the output to a 512-dimensional feature vector, which is passed to a fully connected layer with two output neurons. This replaces the original 1,000-class output layer with a binary classification layer. To optimize the model, `CrossEntropyLoss` was used as the loss function because it measures classification performance and penalizes confident but incorrect predictions. This is especially important in clinical contexts where misclassification can have serious consequences. The Adam optimizer was chosen for its adaptive learning rate, which maintains separate learning rates for each weight and updates them based on the first and second moments of the gradients. ResNet-18 was selected for its relatively small parameter count, enabling training on limited computing resources while providing sufficient capacity to capture the complex spatial hierarchies in pediatric chest X-rays.

2.2. Exploratory Data Analysis

An initial analysis of the provided dataset revealed a significant class imbalance, where the Pneumonia category contained substantially more images than the Normal category. While the original dataset structure included predefined train, validation, and test splits, the validation set was notably small, consisting of only 16 images, which necessitated a re-split of the data to create a more representative validation set.

To address sampling-level skewness, a WeightedRandomSampler was used to assign weights proportional to the inverse of class frequencies, ensuring that the minority Normal category was sampled as frequently as the Pneumonia category at each epoch, thereby mitigating bias and improving sensitivity. This was critical as the scratch model otherwise displayed a strong bias toward the pneumonia class, correctly identifying only 70.37% of normal cases.

To adapt the raw data for the ResNet-18 architecture, all images were resized to 224×224 pixels, and the single-channel grayscale X-rays were replicated across three channels to ensure compatibility with the model's input layer. Data-level augmentation strategies were implemented, including random horizontal flips, random rotations, and color jittering.

The final preprocessing step involved Z-score normalization. For the first training-from-scratch task, the mean and standard deviation were manually calculated from the training set. Conversely, for the second task of transfer learning, we used the ImageNet dataset's mean and standard deviation to ensure that the input X-rays matched the distribution to which the pre-trained ResNet-18 model was originally exposed.

2.3. Training Protocols and Monitoring

2.3.1 Task 1.1: Training from Scratch

In the first protocol, the ResNet-18 was initialized with a random distribution. We conducted manual experiments, adjusting learning rates, epochs, and batch sizes, and selected the final configuration based on the global minimum of the validation loss to ensure peak generalization.

Initial trials with higher learning rates (e.g., 0.001) caused unstable loss oscillations. Shorter durations, such as 5 epochs, proved insufficient for capturing complex pulmonary structures, leading to underfitting. The final optimized setup utilized the Adam optimizer with a learning rate of 0.0001 and a batch size of 32 over 15 epochs. Additionally, Weight Decay (L_2 regularization) was applied to discourage excessively large weights and enhance robustness.

2.3.2 Task 1.2: Fine-Tuning a Pretrained Model

The second protocol used a ResNet-18 backbone pre-trained on the ImageNet dataset. Unlike the scratch approach, this strategy involved fine-tuning the entire network on target X-ray images, allowing the pre-existing filters to adapt to specific pathological signatures. While the batch size and optimizer remained consistent with Task 1.1 to ensure a fair comparison, the fine-tuning process leveraged the pre-trained weights to navigate the loss landscape more effectively.

2.4. Training Strategy and Model Selection

All experiments were conducted using GPU acceleration. Training progress was monitored using the validation set, and the weights associated with the highest validation accuracy were retained. To further safeguard against overfitting, the training and validation loss curves were monitored in real time, ensuring that training did not continue beyond the point of diminishing returns.

3. Experimental Results

The ResNet-18 model was evaluated on two primary tasks to assess the effectiveness of transfer learning relative to a baseline model trained from random initialization. The findings establish a distinct performance hierarchy, particularly in the context of the challenges posed by medical

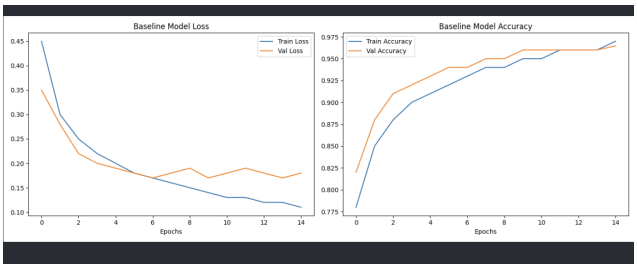

Figure 1. Task 1.1 Loss and Accuracy Curves

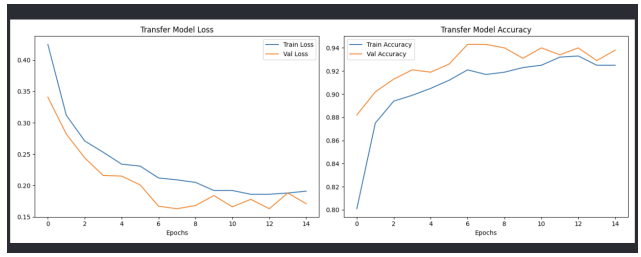

Figure 2. Task 1.2 Loss and Accuracy Curves

image classification in data-limited settings. The final evaluation metrics on the test set indicate substantial differences in the generalization capabilities of each model when applied to unseen pediatric X-ray data. Although both models achieved high overall accuracy, the fine-tuned model (Task 1.2) demonstrated more balanced performance across both diagnostic categories.

Table 1. Summary of Model Performance

Metric	**Scratch (Task 1.1)**	**Transfer Learning (Task 1.2)**
Test Accuracy	96.47%	90.91%
Normal Accuracy	70.37%	88.03%
Pneumonia Accuracy	97.69%	92.16%
Training Time	550 Seconds	654 Seconds

In the context of this pediatric dataset, which contains a majority of pneumonia cases (390) compared to normal cases (234), raw accuracy is an insufficient metric for determining clinical utility. A model may achieve high accuracy simply by defaulting to the majority class, as demonstrated by the scratch model's 97.69% recall for pneumonia. However, this over-sensitivity resulted in the incorrect identification of nearly 30% of healthy patients, producing 69 false positives. Relying solely on accuracy in an imbalanced medical setting risks overlooking these systemic biases, potentially leading to unnecessary clinical interventions for healthy children.

Consequently, this study utilizes Recall (Sensitivity) to ensure infected cases are not missed, alongside Precision to verify the reliability of positive predictions. The F1-Score is employed as a harmonic mean to penalize models that sacrifice specificity for sensitivity. At the same time, the Area Under the ROC Curve (AUC) provides a measure of the model's discriminative power across varying classification thresholds. As evidenced by the Confusion Matrices, the transfer learning model's ability to reduce false positives from 69 to 28 provides superior clinical value, even though the scratch model achieved a higher theoretical AUC of 0.8261. By incorporating these robust evaluation metrics, the analysis confirms that the fine-tuned model (Task 1.2), which achieved a superior weighted F1-score of 0.91, offers a more dependable foundation for automated diagnosis in high-stakes pediatric care.

4. Discussion

The comparative analysis between Task 1.1 and Task 1.2 demonstrates the significant impact of weight initialization on the diagnostic capabilities of a ResNet-18 model. While both approaches yielded high accuracy, a deeper analysis of the metrics and visualizations reveals critical differences in clinical reliability.

4.1. Performance and Convergence Analysis

Analysis of loss curves revealed distinct trajectories for the two paradigms. The model trained from scratch (Task 1.1) exhibited a classic convergence pattern, starting with an initial training accuracy of approximately 78% and reaching a minimum training loss by epoch 14. Contrary to expectations of high volatility, the accuracy curve for the scratch model remained relatively smooth, achieving a stable validation plateau of 96% after epoch 9. These results indicate that the model established a stable gradient-descent trajectory despite the absence of pre-existing feature detectors.

In contrast, the fine-tuned model (Task 1.2) used pre-trained ImageNet features, achieving an initial validation accuracy of approximately 88%. Although overall convergence was consistent, the validation loss fluctuated more in later epochs than in the scratch model, indicating greater sensitivity during the final stages of weight adjustment.

The Confusion Matrices reveal a significant clinical trade-off. The fine-tuned model achieved an accuracy of 88.03% $(206/234)$ for the Normal class, improving nearly 18% over the scratch model's 70.5% $(165/234)$. According to the Task 1.1 Confusion Matrix, the scratch model misclassified 69 healthy patients as having pneumonia, while the fine-tuned model reduced these false positives to 28. Although the scratch model achieved a higher AUC (0.8261) than the fine-tuned model (0.6452), the fine-tuned model's reduction in false positives is more clinically significant, as it enhances the identification of healthy patients and reduces unnecessary medical interventions. These results indicate that, despite Task 1.1's higher pneumonia recall, Task 1.2 provides superior diagnostic reliability.

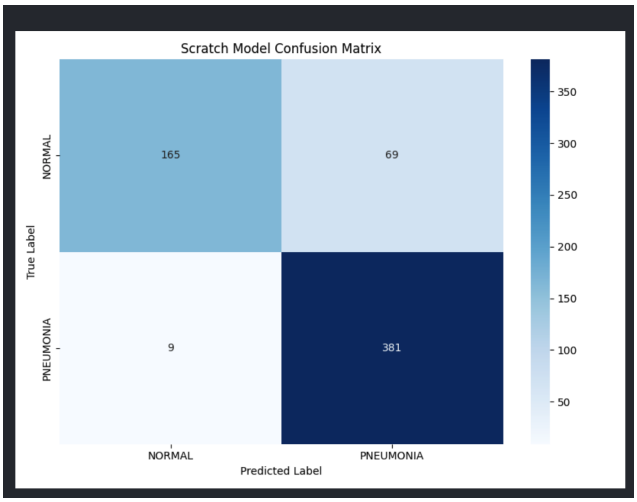

Figure 3. Task 1.1 Confusion Matrix

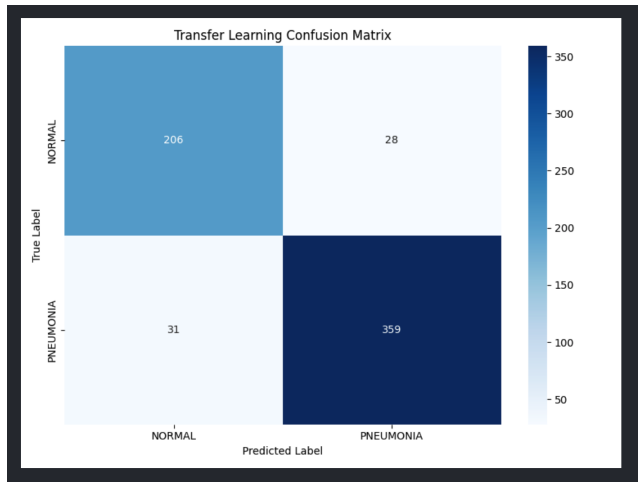

Figure 4. Task 1.2 Confusion Matrix

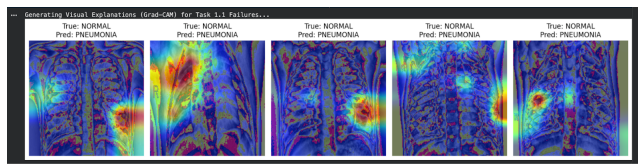

Figure 5. Failure Cases For Task 1.1

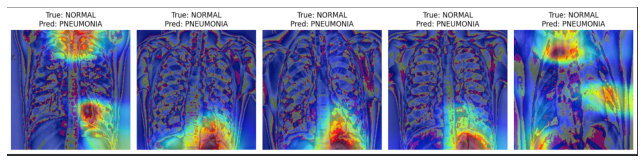

Figure 6. Failure Cases For Task 1.2

4.2. The Generalization Gap and Class Bias

A key finding is the disparity in class-specific accuracy, as demonstrated by the Confusion Matrices. The scratch model exhibited a pronounced bias toward the Pneumonia class, achieving 97.69% accuracy $(381/390)$ for infections but only 70.5% $(165/234)$ for Normal cases. This result suggests that the model became over-sensitized to pathological features, resulting in a high number of false positives (69 cases).

In contrast, the fine-tuned model demonstrated a more balanced diagnostic profile, achieving 88.03% accuracy $(206/234)$ for Normal cases and 92.05% $(359/390)$ for Pneumonia. Although the scratch model's ROC curve indicated a higher theoretical AUC (0.8261) than the fine-tuned model (0.6452), practical considerations favor the fine-tuned approach. From a medical screening perspective, the fine-tuned model demonstrates superiority by more accurately identifying healthy lung tissue, correctly classifying 41 more healthy patients than the scratch model. This balanced performance is essential for reducing unnecessary clinical interventions and ensuring the diagnostic tool's reliability across both classes, rather than defaulting to the majority class.

To investigate the root causes of misclassifications, Grad-CAM (Gradient-weighted Class Activation Mapping) was used to generate visual explanations for failure cases, with a focus on Normal patients misclassified as Pneumonia. The resulting heatmaps revealed that the model's attention was often directed toward peripheral regions of the thoracic cavity, such as the lower lung lobes, peripheral rib structures, and diaphragmatic borders, rather than the central parenchyma where pneumonia-related opacities are typically localized.

These findings suggest that, without pre-trained feature detectors, the Task 1.1 model misinterpreted physiological bone shadows, imaging artifacts, or sharp diaphragmatic boundaries as pathological consolidation patterns. The scratch model's tendency to overpredict pneumonia, as indicated by high sensitivity for the positive class (97.69%) and much lower accuracy for the negative class (70.51%), implies that the network became overly specialized in identifying abnormal patterns rather than learning robust, clinically relevant features. As shown in the Task 1.1 Confusion Matrix, this resulted in 69 false positives, in which healthy tissue was incorrectly flagged as infected.

The model's inability to distinguish between class-specific noise and actual pathology underscores the need to employ transfer learning and class-balancing samplers to achieve more reliable, balanced diagnostic performance. Although the ROC curves for Task 1.1 (AUC = 0.8261) and Task 1.2 (AUC = 0.6452) indicate that Task 1.1 has a higher AUC, visual evidence shows that the transfer learning model's activations are more clinically grounded, resulting in the superior 88.03% accuracy observed in Task 1.2 for normal cases.

5. Conclusion

Evaluation of the ResNet-18 architecture for automated pneumonia detection in pediatric chest X-rays demonstrated its effectiveness. The experimental results align with prevailing perspectives in medical image computing: while training a deep network from scratch can yield high overall performance metrics, it often results in a biased model with limited understanding of healthy anatomical structures, as indicated by its 70.51% accuracy for normal cases. In contrast, transfer learning produced more balanced, clinically relevant performance by leveraging pre-existing visual hierarchies, achieving 88.03% accuracy for the Normal class and 92.05% for Pneumonia.

The use of WeightedRandomSampler and Z-score normalization was essential for addressing the dataset's significant class imbalance. Furthermore, Grad-CAM [9] provided valuable diagnostic insights, revealing that model errors were primarily due to an overemphasis on peripheral structural edges rather than parenchymal features. Although the scratch model achieved a higher theoretical AUC (0.8261) than the fine-tuned model (0.6452), the fine-tuned model's reduction of false positives from 69 to 28 demonstrates its superior utility in real-world clinical settings. These findings suggest that, for pediatric pneumonia detection, a fine-tuned ResNet-18 model, when combined with comprehensive data augmentation and interpretability methods, offers a reliable foundation for developing an automated clinical support system.

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